VISIONINSPECT AI

Milestone 2 – Image Processing, AI Defect Detection and Inspection Workflow

Milestone	Milestone 2
Focus	Implement image acquisition, preprocessing, AI-based anomaly detection and the core inspection workflow.

1. Introduction

Milestone 2 focused on transforming the application foundation created in Milestone 1 into a functional computer-vision inspection system. The main work involved dataset integration, image processing, anomaly detection and connecting AI inference to the inspection interface.

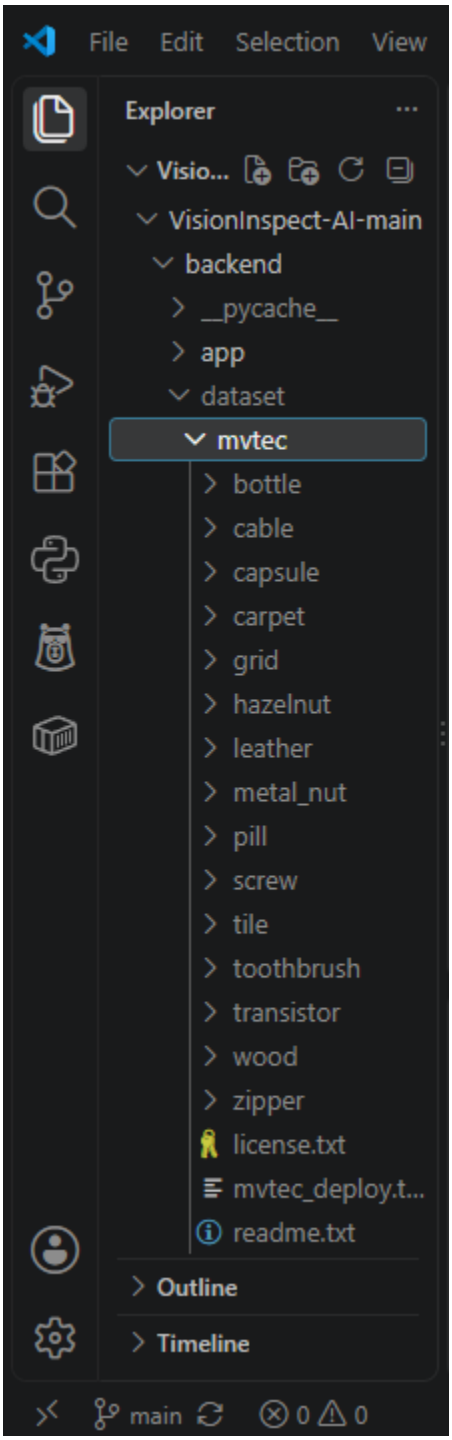
2. Milestone 2 Objectives

- Integrate the MVTec AD dataset for manufacturing defect inspection.
- Implement image upload and validation in the inspection workflow.
- Develop image preprocessing and feature-extraction functionality.
- Implement AI-based anomaly/defect detection.
- Connect the AI service to the FastAPI inspection endpoint.
- Display inspection results through the frontend.
- Store completed inspection results for later history and analytics.

3. Dataset Integration

The MVTec Anomaly Detection dataset was integrated as the reference dataset for product anomaly inspection. The project includes 15 product categories used during evaluation and inspection development.

- Bottle
- Cable
- Capsule
- Carpet
- Grid
- Hazelnut
- Leather
- Metal Nut
- Pill
- Screw
- Tile
- Toothbrush
- Transistor
- Wood
- Zipper



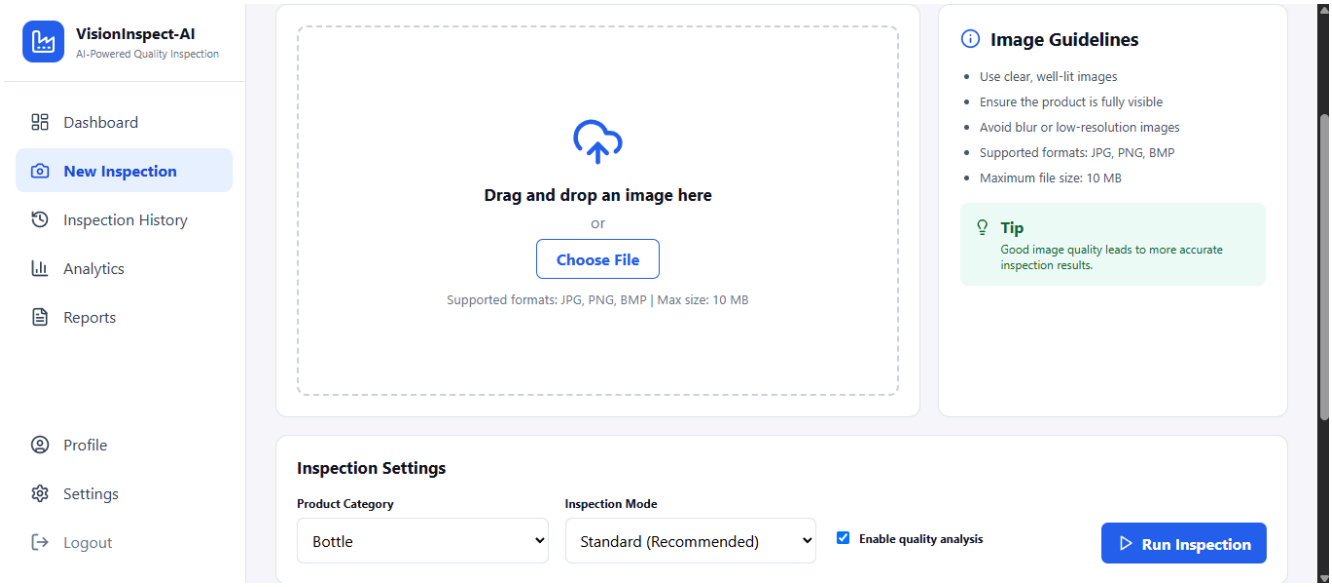
4. Image Acquisition and Validation

A dedicated New Inspection interface was implemented to allow the user to upload an image, select the product category and choose the inspection mode before starting an inspection.

- Drag-and-drop / file-selection workflow.
- Supported image formats include JPG, PNG and BMP.
- Maximum upload size is 10 MB.
- Image-quality guidance is displayed to encourage clear and well-lit images.

- Product category selection is provided before inspection.

New Inspection – Image Upload

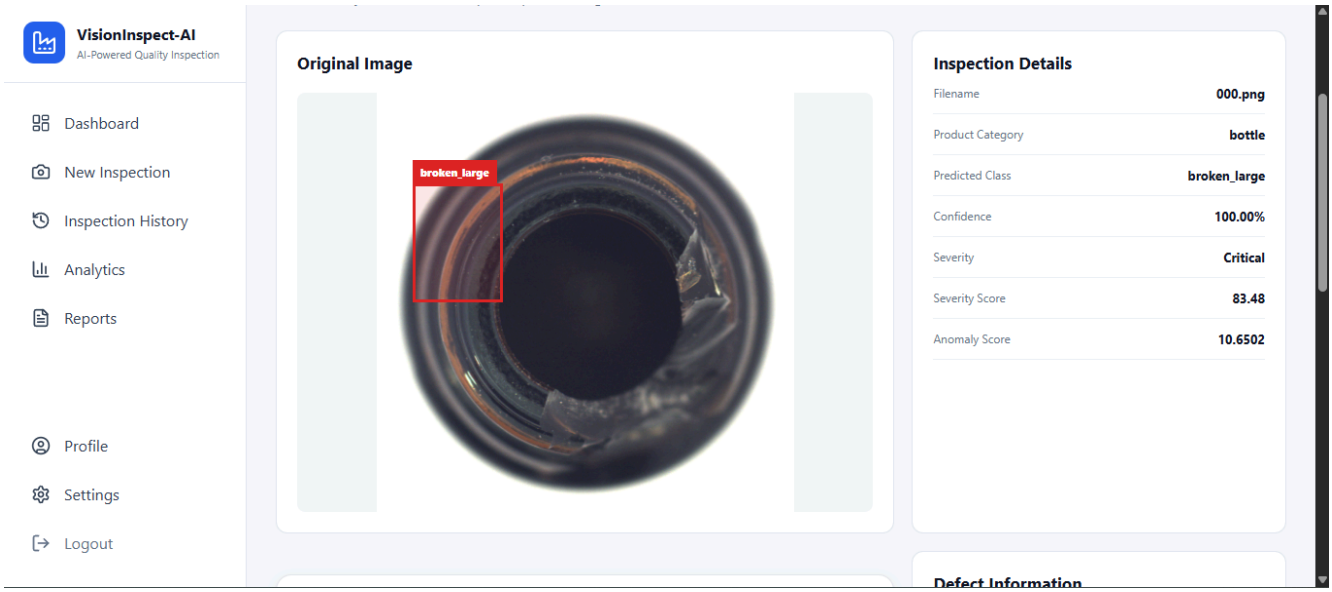


5. Image Processing and Feature Extraction

The anomaly detection service uses a pretrained ResNet18 network as a feature extractor. Image features are compared with reference images from the selected MVTec category to estimate whether the submitted image is normal or anomalous.

- Input image is loaded and prepared for model inference.
- ResNet18 is used to extract deep visual features.
- Reference features are obtained from normal training images.
- Feature similarity/distance is used to derive an anomaly score.
- A category-specific threshold determines the anomaly decision.

AI Processing / Inspection Result:



6. Anomaly Detection Workflow

1. User selects a product category.
2. User uploads an inspection image.
3. The backend validates and receives the image.
4. The image is processed for model inference.
5. ResNet18 extracts visual feature representations.
6. The extracted representation is compared with normal reference images.
7. An anomaly score is calculated.
8. The configured threshold is applied.
9. The inspection result is returned to the frontend.
10. The result is stored in the inspection database.

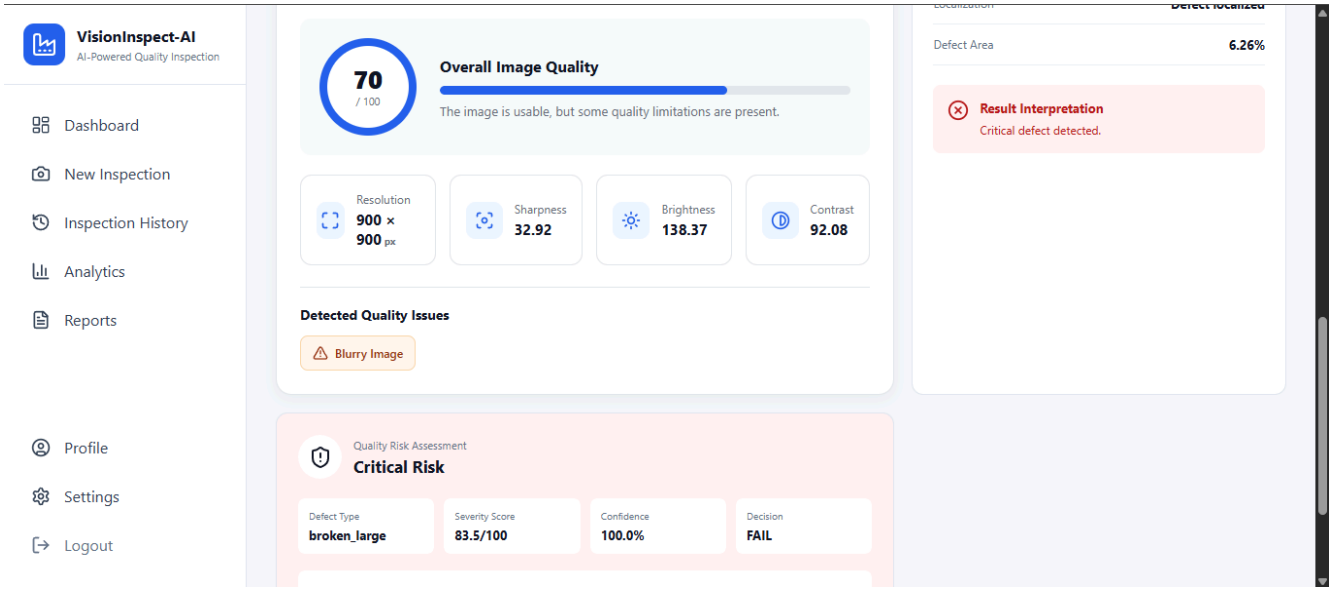
7. Backend Inspection API

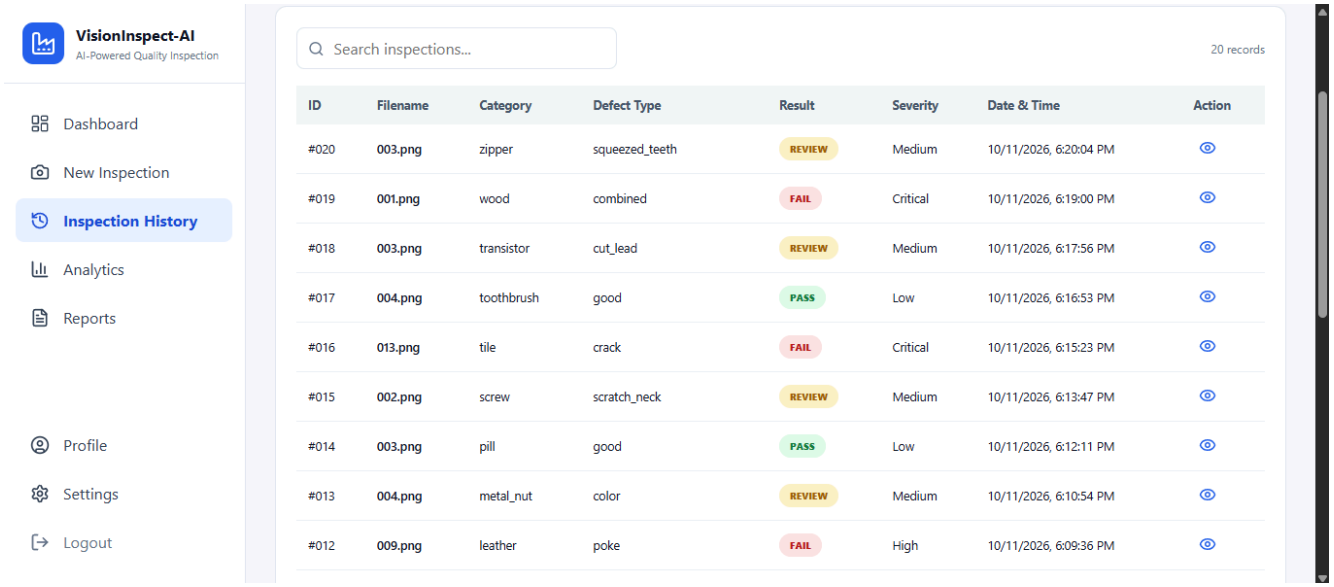
The core inspection functionality was exposed through FastAPI. The inspection route receives the uploaded image and selected category, performs the AI inspection and returns the structured inspection result.

- GET /api/inspection/categories – provides available inspection categories.
- POST /api/inspection/inspect – performs the image inspection.
- Inspection results are persisted for subsequent history and analytics.

8. Inspection Result and Quality Information

The inspection workflow was extended to return information that can be used by the application for quality decisions, including anomaly/defect status and inspection metadata.





9. Milestone 2 Outcome

Milestone 2 converted the initial application foundation into a working AI inspection workflow. The system could accept product images, process them through the computer-vision pipeline, determine anomaly status and store inspection results.

10. Conclusion

Milestone 2 successfully established the core AI-powered visual inspection capability of VisionInspect AI. The application was ready for the next milestone, which focused on classification, severity, quality control and manufacturing analytics.